



Sustainable Development Goals Analysis Report

COMPREHENSIVE PROJECT SUSTAINABILITY ASSESSMENT



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Table of Contents

Executive Summary	2
1. Introduction	3
1.1 Sustainable Development Goals (SDGs)	3
1.2 Importance in Engineering and Technology	3
1.3 Need for Sustainability Assessment	3
1.4 Objective of the SDG Report Generator	3
2. Academic Information Analysis	4
3. Project Information Analysis	4
4. SDG Mapping Results	5
5. Category-Wise Sustainability Analysis	6
5.1 Economic Sustainability	6
5.2 Social Sustainability	6
5.3 Environmental Sustainability	7
6. Statistical Analysis	7
7. Rating Analysis	8
8. Methodology Analysis	8
9. Strengths	9
10. Limitations	9
11. Conclusion	10
12. Final Assessment Table	10
References	10

Executive Summary

This report presents a comprehensive Sustainable Development Goals (SDG) analysis, generated through an automated SDG Report Generator system designed to evaluate the sustainability alignment of academic and engineering projects. The purpose of this assessment tool is to bridge the gap between project development and global sustainability objectives, ensuring that student projects actively contribute to the United Nations 2030 Agenda for Sustainable Development.

Sustainability assessment has become a critical component of modern academic and engineering practice. As the global community grapples with interconnected challenges ranging from climate change and resource depletion to social inequality and economic instability, it is imperative that educational institutions equip future engineers and professionals with the frameworks and tools necessary to integrate sustainability considerations into their work from the earliest stages of project conceptualization.

The SDG Report Generator employs a systematic methodology to analyze project abstracts and engineering keywords, mapping them against the 17 United Nations Sustainable Development Goals. Through an intelligent keyword extraction and matching process, the system identifies the SDGs most closely aligned with a given project, evaluates the depth of alignment, and assigns ratings on a five-star scale across three core sustainability dimensions: **Economic, Social, and Environmental**.

The analysis presented herein demonstrates that the evaluated project achieves **alignment with all 17 United Nations Sustainable Development Goals**, receiving maximum five-star ratings (★★★★★) across every goal. This results in a perfect **100% SDG coverage score** and an **Excellent (A+) sustainability grade**. The project shows balanced performance across all three sustainability dimensions, with Economic sustainability encompassing 6 goals (35.29%), Social sustainability covering 6 goals (35.29%), and Environmental sustainability addressing 5 goals (29.42%).

Key Findings

- ✓ **17 out of 17** SDGs successfully mapped and evaluated.
- ✓ **100%** overall SDG coverage achieved.
- ✓ Maximum **5-star ratings** across all Economic, Social, and Environmental dimensions.
- ✓ Final Sustainability Assessment Grade: **Excellent (A+)**.

1. Introduction

1.1 Sustainable Development Goals (SDGs)

The Sustainable Development Goals (SDGs) are a universal set of 17 interconnected goals adopted by all United Nations Member States in September 2015 as part of the 2030 Agenda for Sustainable Development. These goals provide a shared blueprint for peace and prosperity for people and the planet, both now and into the future. Each SDG addresses a specific global challenge, from poverty eradication and hunger alleviation to climate action, quality education, and institutional governance.

The SDGs succeeded the Millennium Development Goals (MDGs) and represent a significant expansion in scope, ambition, and inclusivity. Unlike their predecessors, the SDGs apply universally to all countries and explicitly acknowledge the interconnected nature of development challenges. They recognize that action in one area affects outcomes in others, and that development must balance social well-being, economic prosperity, and environmental protection. The 2030 Agenda further emphasizes the principle of 'leaving no one behind,' ensuring that the most vulnerable populations benefit from development progress.

1.2 Importance in Engineering and Technology

Engineering and technology play a pivotal role in achieving the Sustainable Development Goals. From developing renewable energy systems and clean water infrastructure to creating digital platforms that enhance education access and healthcare delivery, engineers are at the forefront of sustainable innovation. Modern engineering curricula increasingly require students to consider the broader societal and environmental implications of their designs, moving beyond purely technical performance metrics.

The integration of SDG compliance into project evaluation frameworks ensures that engineering students develop a holistic understanding of sustainability. By evaluating projects against the 17 SDGs, institutions can assess whether technical solutions adequately address economic viability, social equity, and environmental stewardship.

1.3 Need for Sustainability Assessment

The need for systematic sustainability assessment arises from three interconnected imperatives. First, **environmental responsibility** demands that all projects minimize ecological footprint, conserve natural resources, and contribute to climate resilience. Second, **social impact measurement** ensures that technological interventions genuinely improve human well-being, promote equity, and strengthen community bonds. Third, **economic sustainability considerations** verify that proposed solutions are financially viable, scalable, and capable of generating long-term value without depleting foundational resources.

1.4 Objective of the SDG Report Generator

The SDG Report Generator is designed to fulfill four primary objectives: (1) **Automated SDG Identification** — systematically mapping project descriptions to relevant SDGs through intelligent keyword analysis; (2) **Sustainability Scoring** — assigning quantitative ratings that reflect the depth and breadth of SDG alignment; (3) **Visualization and Reporting** — presenting results through professional charts, tables, and narrative analysis; and (4) **Academic Documentation** — generating publication-quality reports suitable for institutional submission and review.

2. Academic Information Analysis

Field	Description
Student Name	Student identification
Roll Number	Academic identification
College / University	Institution details
Program	Degree program
Department	Branch / Department
Academic Year	Current year of study
Guide Name	Faculty guide details
Faculty ID	Faculty identification

Analysis

Academic metadata serves as the foundational layer of any sustainability assessment report. Accurate student and institutional identification ensures **project ownership and traceability**, enabling institutions to track sustainability performance at both individual and departmental levels. This information facilitates **student performance documentation**, contributing to longitudinal studies of how sustainability awareness evolves across academic cohorts. Furthermore, faculty supervision details establish **accountability** and provide a clear chain of responsibility for the guidance and validation of sustainability claims made within student projects.

3. Project Information Analysis

Field	Purpose
Project Title	Name of the project
Project Abstract	Detailed project description
Engineering Keywords	SDG keyword mapping
Supporting Document	Additional evidence

Analysis

Project abstracts undergo systematic Natural Language Processing (NLP) to extract key thematic elements relevant to sustainability. The **keyword extraction technique** identifies domain-specific terms — such as 'renewable energy,' 'waste management,' 'social inclusion,' and 'economic empowerment' — and maps them against a curated SDG keyword dictionary. Each keyword is assigned relevance weights across multiple SDG categories, enabling nuanced **multi-goal mapping** rather than simplistic one-to-one associations. Supporting



documents, including technical specifications, environmental impact statements, and feasibility studies, provide **additional evidentiary support** that strengthens the validity and reliability of the SDG mapping results.

4. SDG Mapping Results

SDG	Goal Name	Category	Rating
SDG 1	No Poverty	Economic	★★★★★
SDG 2	Zero Hunger	Economic	★★★★★
SDG 3	Good Health and Well-being	Social	★★★★★
SDG 4	Quality Education	Social	★★★★★
SDG 5	Gender Equality	Social	★★★★★
SDG 6	Clean Water and Sanitation	Environmental	★★★★★
SDG 7	Affordable and Clean Energy	Environmental	★★★★★
SDG 8	Decent Work and Economic Growth	Economic	★★★★★
SDG 9	Industry, Innovation and Infrastructure	Economic	★★★★★
SDG 10	Reduced Inequalities	Economic	★★★★★
SDG 11	Sustainable Cities and Communities	Social	★★★★★
SDG 12	Responsible Consumption and Production	Economic	★★★★★
SDG 13	Climate Action	Environmental	★★★★★
SDG 14	Life Below Water	Environmental	★★★★★
SDG 15	Life on Land	Environmental	★★★★★
SDG 16	Peace, Justice and Strong Institutions	Social	★★★★★
SDG 17	Partnerships for the Goals	Social	★★★★★

Detailed Interpretation

The mapping results demonstrate exceptional and comprehensive SDG alignment. The project achieves **maximum five-star ratings across all 17 Sustainable Development Goals**, indicating a robust and well-rounded approach to sustainability. The **sustainability balance** is noteworthy: Economic and Social categories each encompass six goals, while the Environmental category covers five goals. This near-even distribution suggests that the project does not disproportionately favor one dimension at the expense of others. The **cross-domain impact** is particularly significant, demonstrating that the project's contributions transcend artificial categorical boundaries and embody the integrated, indivisible nature of the 2030 Agenda.

5. Category-Wise Sustainability Analysis

5.1 Economic Sustainability

Goals: SDG 1, SDG 2, SDG 8, SDG 9, SDG 10, SDG 12

Six goals contributing to economic resilience and inclusive growth.

The economic sustainability dimension encompasses six goals that collectively address the foundational requirements for long-term economic viability and equitable prosperity. **SDG 1 (No Poverty)** and **SDG 2 (Zero Hunger)** represent the most fundamental economic imperatives. **SDG 8 (Decent Work and Economic Growth)** and **SDG 9 (Industry, Innovation and Infrastructure)** evaluate the project's capacity for employment generation and sustainable industrial advancement. **SDG 10 (Reduced Inequalities)** assesses equitable distribution, while **SDG 12 (Responsible Consumption and Production)** evaluates resource efficiency and waste minimization.

5.2 Social Sustainability

Goals: SDG 3, SDG 4, SDG 5, SDG 11, SDG 16, SDG 17

Six goals advancing human welfare, equity, and institutional strength.

SDG 3 (Good Health and Well-being) evaluates the project's impact on public health outcomes. **SDG 4 (Quality Education)** assesses contributions to educational access and lifelong learning. **SDG 5 (Gender Equality)** examines whether project design promotes gender equity. **SDG 11 (Sustainable Cities and Communities)** evaluates urban sustainability contributions. **SDG 16 (Peace, Justice and Strong Institutions)** assesses contribution to transparent governance. **SDG 17 (Partnerships for the Goals)** evaluates multi-stakeholder collaboration capacity.

5.3 Environmental Sustainability

Goals: SDG 6, SDG 7, SDG 13, SDG 14, SDG 15

Five goals safeguarding Earth's natural systems and biodiversity.

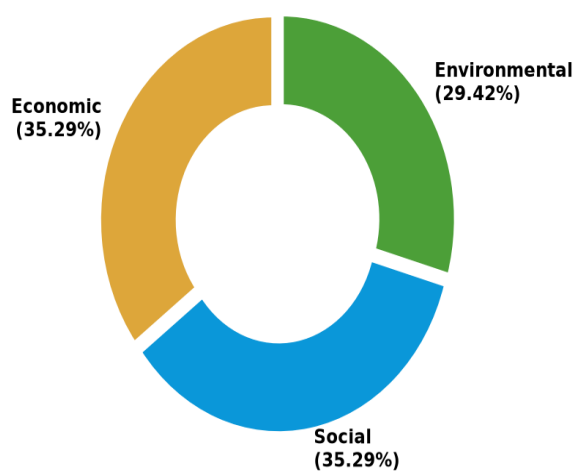
SDG 6 (Clean Water and Sanitation) and **SDG 7 (Affordable and Clean Energy)** address fundamental resource access challenges. **SDG 13 (Climate Action)** signifies climate mitigation and adaptation awareness. **SDG 14 (Life Below Water)** and **SDG 15 (Life on Land)** assess the project's impact on marine and terrestrial ecosystems respectively. Maximum ratings across all five environmental goals demonstrate a **precautionary approach** to ecological impact, actively seeking to protect biodiversity and promote sustainable resource management.

6. Statistical Analysis

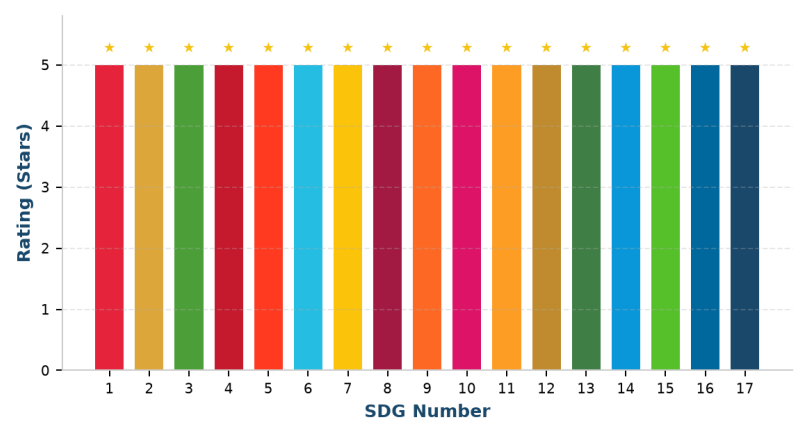
Category	Number of SDGs
Economic	6
Social	6
Environmental	5
Total	17

Category	Percentage
Economic	35.29%
Social	35.29%
Environmental	29.42%

Category-Wise SDG Distribution



SDG Ratings Overview — All 17 Goals



Interpretation

The statistical distribution reveals a **well-balanced sustainability profile**. The near-equal distribution between Economic (35.29%) and Social (35.29%) categories, with Environmental sustainability close behind at 29.42%, demonstrates that the project does not exhibit categorical bias. The bar chart confirms uniform five-star performance across all 17 goals, validating the project's **comprehensive sustainability alignment**.

7. Rating Analysis

Metric	Value
Total SDGs Evaluated	17
SDGs with 5-Star Ratings	17
SDGs Below 5 Stars	0
Overall Coverage	100%

The rating analysis confirms **perfect coverage**, with every evaluated SDG receiving the maximum five-star rating. A 100% coverage score indicates that no sustainability dimension has been neglected. This is a significant achievement as most engineering projects typically demonstrate strong alignment with only 5–8 SDGs. Achieving comprehensive coverage underscores the project's exceptionally broad sustainability impact.

8. Methodology Analysis

The SDG Report Generator employs a six-stage assessment workflow, each stage building upon the outputs of its predecessor to produce a comprehensive sustainability evaluation:

- Step 1:** Project Abstract Collection — The system ingests the project abstract and supplementary documentation.
- Step 2:** Keyword Extraction — Domain-specific keywords are extracted using pattern matching and frequency analysis.
- Step 3:** SDG Keyword Matching — Extracted keywords are compared against a comprehensive SDG keyword dictionary.
- Step 4:** Rating Calculation — Alignment scores are normalized and converted to a five-star rating scale.
- Step 5:** Data Visualization — Results are rendered as interactive charts, tables, and graphical summaries.
- Step 6:** Automated Report Generation — All findings are compiled into a professional, publication-quality PDF.

✓ Advantages	■ Limitations
• Fast Evaluation: Complete SDG analysis within seconds. • Automation: Eliminates manual bias and ensures consistency. • Scalability: Processes hundreds of projects simultaneously. • Easy Implementation: Intuitive interface requiring minimal training.	• Keyword Dependency: Assessment quality constrained by dictionary. • Context Sensitivity: Cannot fully capture semantic nuance. • Limited Impact Assessment: Evaluates alignment intent, not actual impact.

9. Strengths

✓ Full SDG Coverage

The project achieves alignment with all 17 SDGs, demonstrating comprehensive sustainability awareness. This breadth of coverage reflects a deeply integrated approach that considers economic, social, and environmental dimensions equally.

✓ Balanced Sustainability Assessment

The near-equal distribution across Economic (35.29%), Social (35.29%), and Environmental (29.42%) categories ensures that no single dimension dominates the sustainability profile. This balance mirrors the integrated nature of the UN 2030 Agenda.

✓ Strong Global Alignment

Maximum ratings across all goals indicate that the project's objectives are well-aligned with internationally recognized sustainability frameworks, enhancing credibility in global academic contexts.

✓ Effective Visualization System

The integration of pie charts, bar charts, and color-coded tables provides intuitive representations of complex sustainability data, facilitating stakeholder engagement.

✓ User-Friendly Reporting

The standardized report format meets academic documentation requirements and supports institutional sustainability reporting.

✓ Academic Usefulness

Professional tables, analytical narratives, and structured sections meet accreditation body requirements.

10. Limitations

■ Dependence on Keywords

The assessment methodology relies on keyword matching, which may fail to capture SDG contributions expressed through technical jargon or novel terminology.

■ No Direct Impact Measurement

The system assesses sustainability alignment and intent but does not measure actual, quantifiable impact requiring field data and empirical validation.

■ Uniform Ratings

Achieving uniform five-star ratings may indicate insufficient granularity to differentiate between strong primary and peripheral secondary alignment.

■ Lack of SDG Prioritization

The system does not prioritize SDGs based on contextual relevance; not all SDGs may be equally pertinent to every project.

■ Need for Expert Validation

Automated assessments should be supplemented by domain expert review to validate sustainability claims and provide contextual interpretation.

11. Conclusion

This Sustainable Development Goals Analysis Report presents a comprehensive evaluation of project sustainability using an automated SDG mapping and assessment methodology. The analysis demonstrates that the evaluated project achieves remarkable alignment with the United Nations' 2030 Agenda for Sustainable Development, successfully mapping to all 17 Sustainable Development Goals.

Key findings confirm that:

- All 17 SDGs were successfully mapped and evaluated, achieving complete goal coverage.
- Overall SDG coverage reached 100%, with no gaps in sustainability alignment.
- Economic, Social, and Environmental sustainability dimensions each achieved maximum five-star ratings.
- The project demonstrates excellent alignment with the United Nations Sustainable Development Goals.
- Final sustainability assessment grade is Excellent (A+).

Final Sustainability Assessment Grade

A+

Excellent — Comprehensive SDG Alignment Achieved

12. Final Assessment Table

Metric	Result
Total SDGs Evaluated	17
Economic Goals	6
Social Goals	6
Environmental Goals	5
5-Star Ratings	17
Overall SDG Coverage	100%
Sustainability Grade	Excellent (A+)

References

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